

Drone Technology

: Exploring the Parts and Flight Mechanism

A. Warming Up

1. What is the purpose of drone technology?

드론의 목적은 정해진 임무를 수행하는 것임. 다양한 산업 분야에 활용되고 있음.

2. What effects can drone technology have on other industries?

물류업, 농업, 건설업, 방송업 등.

3. What are your concerns about the future of drone technology?

개인정보 침해, 안전사고, 불법 비행 등.

B. Key Expressions

1. We will **delve into** the different parts of a traditional drone and examine how they work together to achieve flight.

>>> Science & Technology Reading

- (A) Through the flight controller (B) Via the propellers
(C) Using the batteries (D) Through the Transmitter and Receiver

5. What is the flight controller's role in a drone?

- (A) Generating lift (B) Regulating motor speed
(C) Processing pilot commands (D) Holding the frame together

6. How can quadcopters counteract torque and achieve stable hovering?

- (A) By adjusting the power to the propellers
(B) By changing the frame size
(C) By increasing battery capacity
(D) By using retractable landing gear

E. Grammar Checkup

Choose one that can best complete the following sentences.

1. We will delve into the different parts of a traditional drone and examine _____ together to achieve flight.

- (A) how they work (B) how do they work
(C) they how work (D) they work how

2. A drone consists of several essential components, each _____ a crucial role in its operation.

- (A) play (B) plays
(C) played (D) playing

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D. Comprehension Checkup

Choose the best answer to the following questions.

1. What are the motors responsible for in a drone?

- (A) Generating lift
(B) Holding the frame together
(C) Controlling the flight controller
(D) Propelling the drone by spinning the propellers

2. Why is it important to select the right propellers for a drone?

- (A) They determine the frame size
(B) They provide structural stability
(C) They generate lift
(D) They affect the drone's stability and maneuverability

3. What is the role of the Electronic Speed Controller (ESC)?

- (A) Generating lift (B) Regulating motor speed
(C) Sending control signals (D) Processing flight data

4. How does communication between the pilot and the drone occur?

3. The frame can have different arm variations, such as tri, quad, hex, or oct, _____ determine the number of propellers the drone will have.

- (A) as (B) but
(C) that (D) which

4. _____ the right propellers contributes to the overall stability and maneuverability of the drone.

- (A) Select (B) Selected
(C) Selecting (D) Being selected

5. As battery capacity increases, _____.

- (A) does so its weight (B) does its weight so
(C) so its weight does (D) so does its weight

6. The number of channels _____ in a transmitter depends on the drone's complexity and desired functionalities.

- (A) require (B) requires
(C) required (D) requiring

F. Vocabulary Checkup

Choose one that can best complete the following sentences.

1. Drones can _____ from autonomous military-grade machines used for critical missions to consumer-friendly remote-controlled devices that enthusiasts fly in



parks.

- (A) assign (B) ☒ enthuse
(C) range (D) supply

2. The motors, positioned at the end of each arm, are responsible for propelling the drone by _____ the attached propellers.

- (A) changing (B) dialing
(C) rolling (D) ☒ spinning

3. Selecting the right propellers contributes to the overall stability and _____ of the drone.

- (A) autonomy (B) ☒ maneuverability
(C) propelling (D) versatility

4. The Electronic Speed Controller (ESC) plays a _____ role in controlling the drone's motors.

- (A) ☒ critical (B) drastic
(C) severe (D) ultimate

5. Communication between the pilot and the drone is _____ by the Transmitter and Receiver.

- (A) annulled (B) ☒ facilitated
(C) torqued (D) required

6. By _____ the power distribution to the propellers, quadcopters can execute forward, backward, or sideward movements.